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PROJECTS

From problems to applications.

The practice questions help you strengthen individual Python concepts.

These projects take the next step: applying those concepts to build complete applications.

ALL PROJECTS ARE MANDATORY.

You are required to work on *4 projects* and submit them within the due date and complete the 6 other and submit them before the program ends, as per your ease and time flexibility.

Plan your work across the complete project list and submit them accordingly.

PROJECT 01 — STUDENT MANAGEMENT SYSTEM

Suggested Features

- Add students
- Display students
- Search students
- Update student records
- Delete students
- Calculate average marks
- Save records

PROJECT 02 — LIBRARY MANAGEMENT SYSTEM

Suggested Features

- Add books
- Search books
- Issue books
- Return books
- Display available books
- Save records

PROJECT 03 — BANK ACCOUNT MANAGEMENT SYSTEM

Suggested Features

- Create accounts
- Deposit money
- Withdraw money
- Check balance
- Maintain transaction history
- Save data

PROJECT 04 — EMPLOYEE PAYROLL SYSTEM

Suggested Features

- Add employees
- Calculate salary
- Generate payslips
- Search employees
- Save payroll data

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PROJECT 05 — INVENTORY MANAGEMENT SYSTEM

Suggested Features

- Add products
- Update products
- Delete products
- Search products
- Generate inventory reports

PROJECT 06 — QUIZ MANAGEMENT SYSTEM

Suggested Features

- Load questions from a file
- Conduct a quiz
- Calculate score
- Save results

PROJECT 07 — HOSPITAL PATIENT RECORD SYSTEM

Suggested Features

- Register patients
- Search patients
- Update patient records
- Delete patient records
- Save records

PROJECT 08 — VEHICLE RENTAL MANAGEMENT SYSTEM

Suggested Features

- Add vehicles
- Rent vehicles
- Return vehicles
- Calculate rental charges
- Save rental records

PROJECT 09 — EXPENSE TRACKER

Suggested Features

- Record income
- Record expenses
- Categorize transactions
- Generate monthly summaries
- Search records
- Save records

PROJECT 10 — COURSE REGISTRATION SYSTEM

Suggested Features

- Add students
- Add courses
- Register students for courses
- Drop courses
- Display registrations
- Save data

EXPECTED OOP CONCEPTS

Every project must demonstrate the required programming concepts meaningfully.

01. Classes & Objects

Use *at least 3 classes*.

02. Encapsulation

Use private/protected attributes with getters/setters.

03. Inheritance

Implement *at least one derived class*.

04. Abstraction

Use at least one abstract base class using the abc module.

05. File Handling

Read and write data using text or CSV files.

06. Exception Handling

Handle invalid input and file-related errors appropriately.

SUBMISSION REQUIREMENTS

For each project, submit:

01. Python Source Code

Submit your Python source code as .py files.

02. Sample Data Files

Include the required sample data files in .txt or .csv format.

03. README

Include a README explaining:

- What the project does
- How the project works
- How to run the project
- The classes used
- The main features implemented

04. Console Output Screenshots

Include screenshots demonstrating the key operations of your application.

BEFORE YOU SUBMIT

Check that:

☐ The project runs.

☐ The required features are implemented.

☐ The code has been tested.

☐ Invalid inputs are handled.

☐ File operations work correctly.

☐ The required OOP concepts are implemented.

☐ The source code is organised.

☐ The README is included.

☐ The required screenshots are included.

☐ Your submission is made before the due date.

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THE STANDARD

Do not build just to complete the list.

Build so the work can be understood, tested, reviewed, and verified.

The project should be something you can open, run, explain, and defend.

That is the point of doing the work.

Clarity over chaos.
Action over hesitation.
Progress over perfection.

Stop consuming. Start doing.